COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/17/2026	**DATE SUBMITTED:**	07/17/2026
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ACTIVITY NO. 2.1 **ARRAYS POINTERS AND DYNAMIC MEMORY ALLOCATION**

1. PROCEDURE OUTPUT

What procedures did you perform?

5. Procedure

ILO A: Implement static and dynamic memory allocation & ILO B: Create dynamically allocated objects using pointers and arrays

In this activity, we will explore static and dynamic memory allocation through by utilizing arrays and other components mentioned beforehand, this will be for the implementation of a student list with the students' names and age. To begin, we will include pertinent libraries for input/output stream and strings (which we will be using in this activity).

```
#include <iostream>
#include <string.h>
```

Now we will create the student class with private attributes `studentName` and `studentAge`.

```
class Student{
private:
    std::string studentName;
    int studentAge;
```

Then, as discussed beforehand, we have to define the constructor and the big three: the destructor, copy constructor and copy assignment operator. We will assign default values for the parameters of our constructor in the event that it is uninitialized.

```
public:
    //constructor
    Student(std::string newName ="John Doe", int newAge=18){
        studentName = std::move(newName);
        studentAge = newAge;
        std::cout << "Constructor Called." << std::endl;
    };
```

The big three is then defined. We will add an output stream for each to observe when each of the following functions are implicitly or explicitly called.

```
    //deconstructor
    ~Student(){
        std::cout << "Destructor Called." << std::endl;
    }

    //Copy Constructor
    Student(const Student &copyStudent){
        std::cout << "Copy Constructor Called" << std::endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    //Display Attributes
    void printDetails(){
        std::cout << this->studentName << " " << this->studentAge << std::endl;
    }

};
```

The driver program is now to be defined. We want to utilize the main function to simply show the static and dynamic allocation. However, we will explore the initial goal of creating instances of our Student class.

```
int main() {
    Student student1("Roman", 28);
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;
}
```

Run the code and show the output. Include the screenshot in table 2-1 followed by your observation.
Now, modify the driver program so that we have the array size of 5, an array of Student objects, the list of students' names and their age.

```
int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};
```

```
        return 0;
    }
```

Run the code with the modified driver function and show the output. Include the screenshot in table 2-2 followed by your observation.

We have so far created static memory allocation through the use of the arrays. We will now dynamically allocate instances of the student class and store the newly created objects in the locations pointed to by our array.

```
int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    for(int i = 0; i < j; i++){ //loop A
        Student *ptr = new Student(namesList[i], ageList[i]);
        studentList[i] = *ptr;
    }

    for(int i = 0; i < j; i++){ //loop B
        studentList[i].printDetails();
    }

    return 0;
}
```

Discuss what is done by loop A and loop B in table 2-3. Additionally, discuss the output and whether the functions are working as intended. If any corrections were made, further provide your modification and analysis in table 2-4.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO A & B INPUT

```cpp
#include <iostream>
#include <string>

using namespace std;

class Student {
private:
    string studentName;
    int studentAge;

public:
    // Constructor
    Student(string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;
        cout << "Constructor Called." << endl;
    }

    // Destructor
    ~Student() {
        cout << "Destructor Called." << endl;
    }

    // Copy Constructor
    Student(const Student& copyStudent) {
        cout << "Copy Constructor Called." << endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    // Copy Assignment Operator
    Student& operator=(const Student& copyStudent) {
        cout << "Copy Assignment Operator Called." << endl;

        if (this != &copyStudent) {
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
```

```
40          return *this;
41      }
42
43      // Display Details
44      void printDetails() {
45          cout << studentName << " " << studentAge << endl;
46      }
47  };
48
49  int main() {
50
51      const size_t j = 5;
52
53      Student studentList[j];
54
55      string namesList[j] = {"KEITH", "NICOLE", "BONDOC", "SILVOZA", "STAR"};
56      int ageList[j] = {15, 16, 18, 19, 16};
57
58      for (int i = 0; i < j; i++) {
59          Student* ptr = new Student(namesList[i], ageList[i]);
60          studentList[i] = *ptr;
61          delete ptr;
62      }
63
64      cout << "\nStudent List\n";
65      cout << "------------" << endl;
66
67      for (int i = 0; i < j; i++) {
68          studentList[i].printDetails();
69      }
```

OUPUT

```
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA>
--stdin=Microsoft-MIEngine-In-rgndw0gz.kvq' '--stdout=
hw1wq.1ku' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.

Student List
------------
KEITH 15
NICOLE 16
BONDOC 18
SILVOZA 19
STAR 16
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA>
```

Table 2-1. Initial Driver Program

SCREENSHOT	

SW_2 > NY.CPP > ...

```cpp
#include <iostream>
#include <string>

using namespace std;

class Student {
private:
    string studentName;
    int studentAge;

public:
    // Constructor
    Student(string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;
        cout << "Constructor Called." << endl;
    }

    // Destructor
    ~Student() {
        cout << "Destructor Called." << endl;
    }

    // Copy Constructor
    Student(const Student& copyStudent) {
        cout << "Copy Constructor Called." << endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    // Copy Assignment Operator
    Student& operator=(const Student& copyStudent) {
        cout << "Copy Assignment Operator Called." << endl;

        if (this != &copyStudent) {
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
        return *this;
    }

    // Display Student Details
    void printDetails() {
        cout << studentName << " " << studentAge << endl;
    }
};

int main() {

    Student student1("Roman", 28);
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;
}
```

	PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA> --stdin=Microsoft-MIEngine-In-q125lih1.qx1' '--stdout=M alqiw.fxl' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--i Constructor Called. Copy Constructor Called. Constructor Called. Copy Assignment Operator Called. Destructor Called. Destructor Called. Destructor Called. PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA>
OBSERVATION	The activity showed how constructors, copy constructors, assignment operators, and destructors work in C++. I learned how objects are created, copied, and deleted automatically by the program.

Table 2-2. Modified Driver Program with Student Lists

SCREENSHOT

```
#include <iostream>
#include <string>

using namespace std;

class Student {
private:
    string studentName;
    int studentAge;

public:
    // Constructor
    Student(string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;
        cout << "Constructor Called." << endl;
    }

    // Destructor
    ~Student() {
        cout << "Destructor Called." << endl;
    }

    // Copy Constructor
    Student(const Student& copyStudent) {
        cout << "Copy Constructor Called." << endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    // Copy Assignment Operator
    Student& operator=(const Student& copyStudent) {
        cout << "Copy Assignment Operator Called." << endl;

        if (this != &copyStudent) {
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
```

```
        return *this;
    }

    // Display Student Details
    void printDetails() {
        cout << studentName << " " << studentAge << endl;
    }
};

int main() {

    const size_t j = 5;

    Student studentList[j] = {};

    string namesList[j] = {
        "Keith",
        "Nicole",
        "Bondoc",
        "Silvoza",
        "Star"
    };

    int ageList[j] = {
        18,
        19,
        18,
        20,
        19
    };

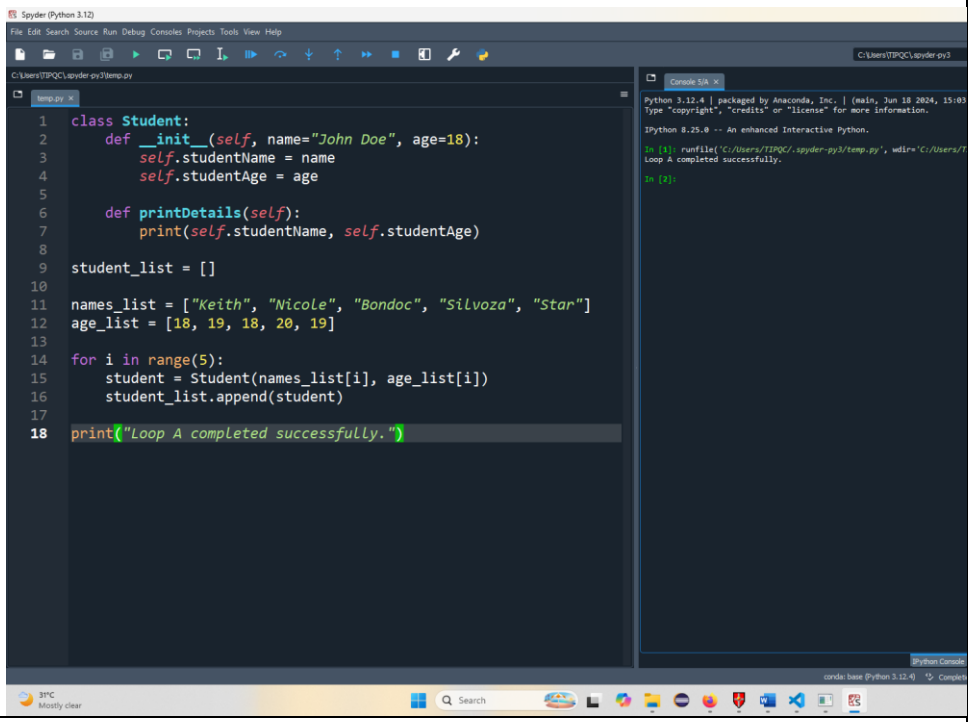
    // Display the student names and ages
    cout << "\nStudent List\n";
    cout << "\nStudent List\n";
    cout << "------------" << endl;

    for (int i = 0; i < j; i++) {
        cout << namesList[i] << " " << ageList[i] << endl;
    }

    return 0;
}
```

	PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA> --stdin=Microsoft-MIEngine-In-a4wtixb2.5lz' '--stdout=y3azc.zcd' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '-- Constructor Called. Constructor Called. Constructor Called. Constructor Called. Constructor Called. Student List ------------ Keith 18 Nicole 19 Bondoc 18 Silvoza 20 Star 19 Destructor Called. Destructor Called. Destructor Called. Destructor Called. Destructor Called. PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-SILVOZA>
OBSERVATION	The modified C++ program successfully created and displayed a list of students using constructors and object initialization. The output showed that the student names and ages were stored and displayed correctly, while the constructor and destructor functions were also executed properly. This demonstrates how classes, objects, and memory handling work in C++.

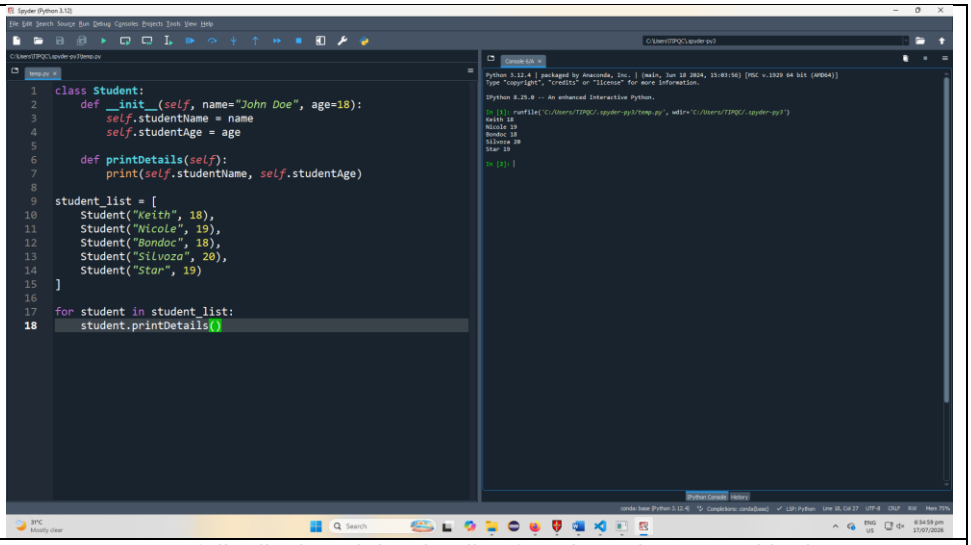
Table 2-3. Final Driver Program

LOOP A	
OBSERVATION	The program successfully displayed the student information using Loop B. Each student object was accessed, and the printDetails() function correctly showed the name and age of every student. The output confirms that the stored data was retrieved properly.

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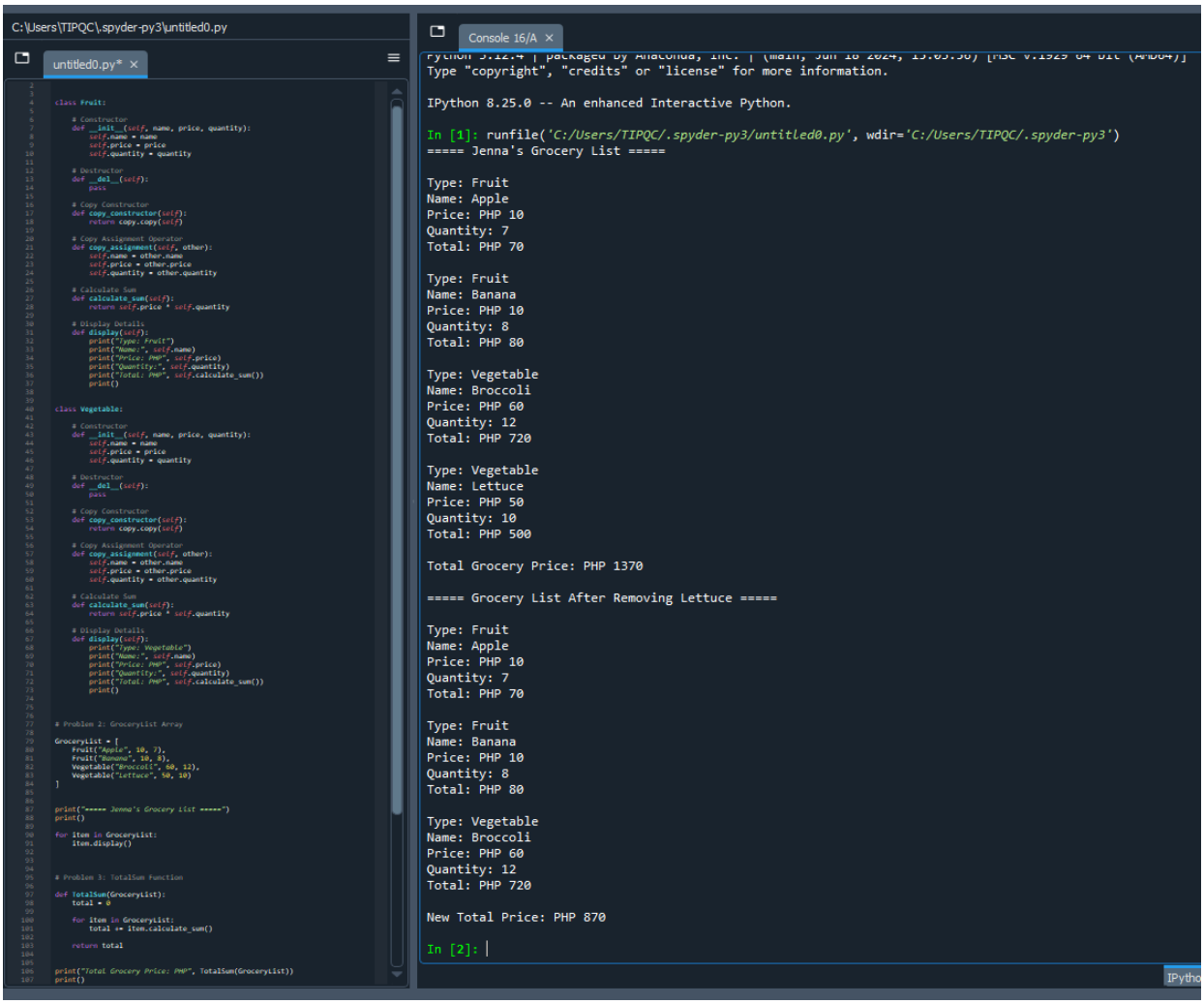
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LOOP B	
OBSERVATION	Loop B successfully displayed the details of each student stored in the studentList. The loop accessed every student object and called the printDetails() function to show their names and ages. The output confirms that the student objects were properly created and stored, and the function worked as intended.
OUTPUT	Console 6/A Python 3.12.4 | packaged by Anaconda, Inc. | (main, Jun 18 2024, 15:03:56) [MSC v.1929 64 bit (AMD64)] Type "copyright", "credits" or "license" for more information. IPython 8.25.0 -- An enhanced Interactive Python. In [1]: runfile('C:/Users/TIPQC/.spyder-py3/temp.py', wdir='C:/Users/TIPQC/.spyder-py3') Keith 18 Nicole 19 Bondoc 18 Silvoza 20 Star 19
OBSERVATION	The output shows the correct list of student names and ages: Keith, Nicole, Bondoc, Silvoza, and Star. This confirms that the program successfully retrieved and displayed the stored data from the student objects.

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-SILVILO C



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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: CHATGPT AND PYTHON VISUALIZER

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: ChatGPT was used for code assistance, explaining programming concepts, identifying errors, and improving the structure of my code. Python Visualizer was used to help visualize program execution, understand the flow of algorithms, and observe how variables and processes change while the code runs.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: AI tools were only used as learning and support resources. They helped me understand, debug, and improve my code, but they did not replace my own

analysis, problem-solving, or completion of the required activity. I reviewed and modified the output to ensure that the final work reflects my own understanding.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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